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Skill Week 12

1. To Change Directories, Validate Paths, Update Working Directory, Handle Errors, Maintain P



```
 UW PICO 5.09                                File: 60.c

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>
#include <errno.h>

#define MAX_PATH 1024

int main() {
    char current_dir[MAX_PATH];
    char previous_dir[MAX_PATH];
    char input[MAX_PATH];

    /* Get initial working directory */
    if (getcwd(current_dir, sizeof(current_dir)) == NULL) {
        perror("getcwd");
        return 1;
    }

    strcpy(previous_dir, current_dir);

    printf("=== Directory Navigation Program ===\n");
    printf("Initial Directory: %s\n", current_dir);

    while (1) {
        printf("\nCurrent Directory: %s\n", current_dir);
        printf("Enter directory path (or 'prev' / 'pwd' / 'exit'): ");

        if (fgets(input, sizeof(input), stdin) == NULL) {
            break;
        }

        /* Remove newline */
        input[strcspn(input, "\n")] = '\0';

        /* Exit */
        if (strcmp(input, "exit") == 0) {
            printf("Exiting program...\n");
            break;
        }

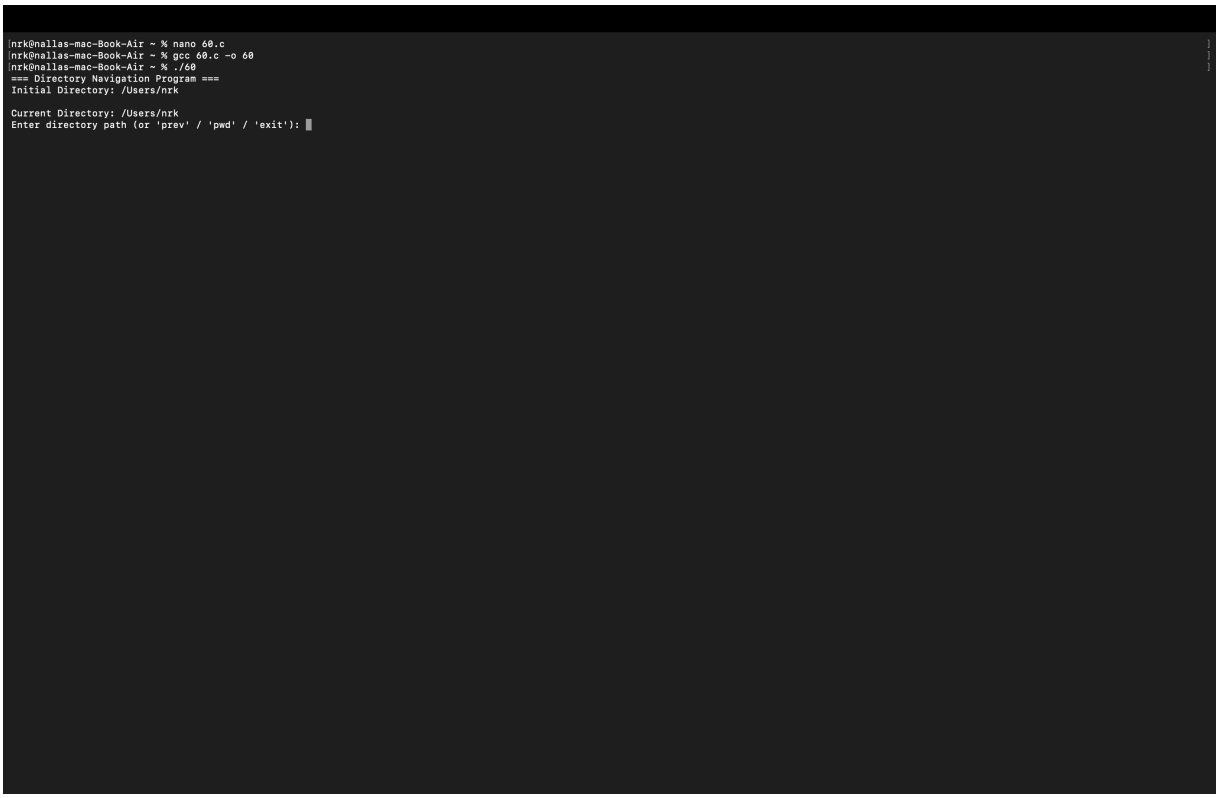
        /* Display current directory */
        if (strcmp(input, "pwd") == 0) {
            printf("Working Directory: %s\n", current_dir);
            continue;
        }

        /* Go to previous directory */
        if (strcmp(input, "prev") == 0) {
            if (chdir(previous_dir) == -1) {
                perror("Cannot change to previous directory");
                continue;
            }

            /* Swap current and previous directories */
            char temp[MAX_PATH];

            strcpy(temp, current_dir);

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text    ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text  ^T To Spell
```



```
nrk@nallas-mac-Book-Air ~ % nano 60.c
nrk@nallas-mac-Book-Air ~ % gcc 60.c -o 60
nrk@nallas-mac-Book-Air ~ % ./60
=== Directory Navigation Program ===
Initial Directory: /Users/nrk

Current Directory: /Users/nrk
Enter directory path (or 'prev' / 'pwd' / 'exit'):
```

2. To Retrieve Current Directory, Display Path, Process Exit Requests, Cleanup Resources, Save State, Verify Correct Behavior

```
 UW PICO 5.09                                File: kk.c

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>

#define MAX_INPUT 100

int main(void)
{
    char current_dir[PATH_MAX];
    char input[MAX_INPUT];

    printf("===== Directory and Exit Handler =====\n");

    while (1)
    {
        /* Retrieve current working directory */
        if (getcwd(current_dir, sizeof(current_dir)) == NULL)
        {
            perror("getcwd");
            return 1;
        }

        /* Display current directory */
        printf("\nCurrent Directory: %s\n", current_dir);
        printf("Enter command (pwd, cd, exit): ");

        if (fgets(input, sizeof(input), stdin) == NULL)
        {
            printf("\nInput error or EOF detected.\n");
            break;
        }

        /* Remove newline */
        input[strcspn(input, "\n")] = '\0';

        /* Handle empty input */
        if (strlen(input) == 0)
        {
            continue;
        }

        /* Display current directory */
        if (strcmp(input, "pwd") == 0)
        {
            printf("Working Directory: %s\n", current_dir);
        }

        /* Exit request */
        else if (strcmp(input, "exit") == 0)
        {
            printf("\nExit request received.\n");

            /* Save current state */
            printf("Saving state: %s\n", current_dir);

            /* Cleanup resources */
            printf("Cleaning up resources...\n");

^G Get Help  ^O WriteOut  ^R Read File  ^Y Prev Pg  ^K Cut Text  ^C Cur Pos
^X Exit      ^J Justify   ^W Where is   ^V Next Pg  ^U UnCut Text ^T To Spell
```

OUTPUT:

```
nrk@nallas-mac-Book-Air ~ % nano kk.c
nrk@nallas-mac-Book-Air ~ % gcc kk.c -o kk
kk.c:7:9: warning: 'MAX_INPUT' macro redefined [-Wmacro-redefined]
    7 | #define MAX_INPUT 100
      |         ^
/Library/Developer/CommandLineTools/SDKs/MacOSX.sdk/usr/include/sys/syslimits.h:90:9: note: previous definition is here
   90 | #define MAX_INPUT               1024    /* max bytes in terminal input */
      |         ^
1 warning generated.
nrk@nallas-mac-Book-Air ~ % ./kk
===== Directory and Exit Handler =====

Current Directory: /Users/nrk
Enter command (pwd, cd, exit): exit

Exit request received.
Saving state: /Users/nrk
Cleaning up resources...
Shell exited successfully.
nrk@nallas-mac-Book-Air ~ %
```

3. To Parse Export Syntax, Update Environment Variables, Validate Names, Handle Existing Variables, Export to Child Processes, Test Functionality



```
  UW PICO 5.09                                File: hhh.c

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>
#include <ctype.h>

#define MAX_INPUT 200
#define MAX_NAME 100
#define MAX_VALUE 100

/* Validate environment variable name */
int valid_name(const char *name)
{
    int i;

    if (name == NULL || name[0] == '\0')
        return 0;

    /* First character must be letter or underscore */
    if (!isalpha((unsigned char)name[0]) && name[0] != '_')
        return 0;

    /* Remaining characters can be letters, digits, or underscore */
    for (i = 1; name[i] != '\0'; i++)
    {
        if (!isalnum((unsigned char)name[i]) && name[i] != '_')
            return 0;
    }

    return 1;
}

/* Process export command */
void process_export(char *input)
{
    char *argument;
    char *equal;
    char name[MAX_NAME];
    char value[MAX_VALUE];
    size_t name_length;

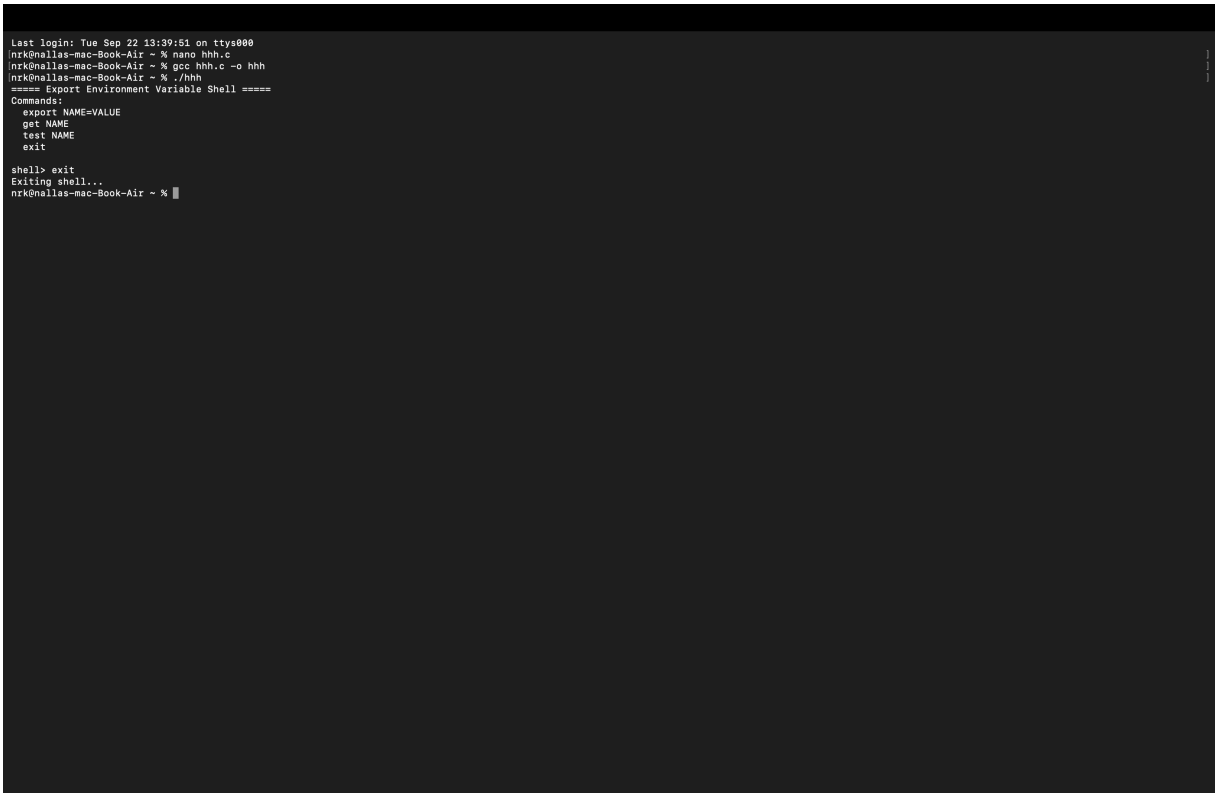
    /* Skip "export" */
    argument = input + 6;

    /* Remove leading spaces */
    while (*argument == ' ' || *argument == '\t')
        argument++;

    /* Check missing argument */
    if (*argument == '\0')
    {
        printf("Usage: export NAME=VALUE\n");
        return;
    }

    /* Find '=' */
    equal = strchr(argument, '=');

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text      ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text    ^T To Spell
```



```
Last login: Tue Sep 22 13:39:51 on ttys000
nrk@nallas-mac-Book-Air ~ % nano hhh.c
nrk@nallas-mac-Book-Air ~ % gcc hhh.c -o hhh
nrk@nallas-mac-Book-Air ~ % ./hhh
===== Export Environment Variable Shell =====
Commands:
  export NAME=VALUE
  get NAME
  test NAME
  exit

shell> exit
Exiting shell...
nrk@nallas-mac-Book-Air ~ %
```